Test #8

1) Each of these 5 clues is a true statement

Clue 1	George, Wendy, and Juanita did not bring their lunches.
Clue 2	George ate the last piece of candy if and only if Juanita brought her lunch.
Clue 3	If Wendy did not bring her lunch, then Juanita did not eat the last piece of candy.
Clue 4	If Dashon ate the last piece of candy, then Wendy brought her lunch.
Clue 5	Only one of these people ate the last piece of candy.

Use these clues to determine who ate the last piece of candy.

A. Dashon
B. George
C. Juanita
D. Wendy

2) Which of the following is $3(4 - x^2) + 10$ in simplified form?

A. $22 - x^2$
B. $22 - 3x^2$
C. $12 - x^2$
D. $12 - 3x^2$

3) Mr. Brady's 9th grade science class grows a type of bacterium in a petri dish. The table shows the number of bacteria in the dish at different times. If the pattern continues, how many bacteria will be in the dish after 10 hours?

Time in hours	Number of bacteria
0123	1248

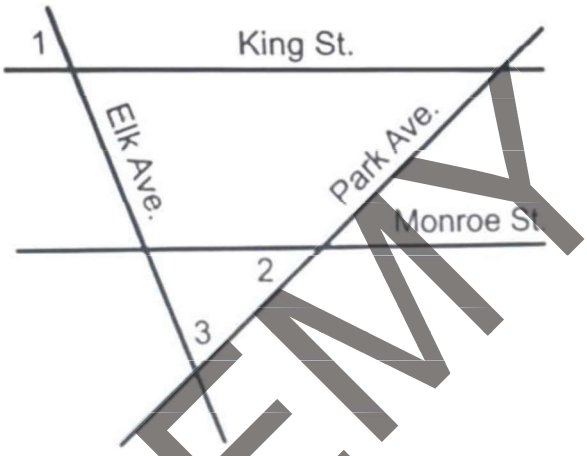
A. 256
B. 512
C. 1,024
D. 2,048

3D ACADEMY

Test #8

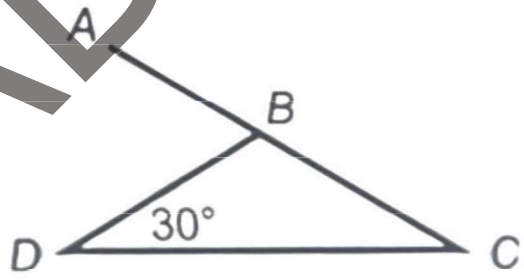
4) This map shows the intersections of four roads. King Street and Monroe Street are parallel. Elk Avenue intersects King Street such that $m\angle 1 = 71°$. Park Avenue intersects Monroe Street such that $m\angle 2 = 43°$. what is $m\angle 3$?

A. $38°$
B. $56°$
C. $66°$
D. $94°$



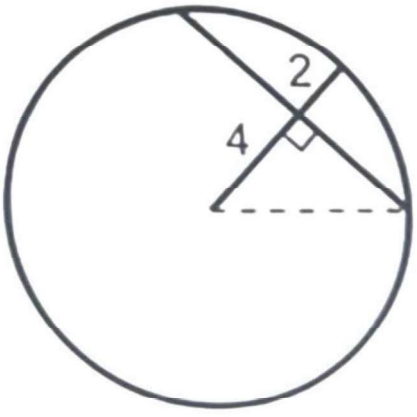
5) In $\Delta\, BCD$, $m\angle BDC = 30°$, if the measure of exterior $\angle ABD$ is twice $m\angle DCB$ what is $m\angle ABD$?

A. $30°$
B. $60°$
C. $105°$
D. $120°$



6) The chord of this circle is perpendicular to a radius of the circle. The chord divides the radius into segment 2cm and 4cm long. What is the length, in centimeters, of the chord?

A. $2\sqrt{5}$
B. $2\sqrt{13}$
C. $4\sqrt{5}$
D. 10



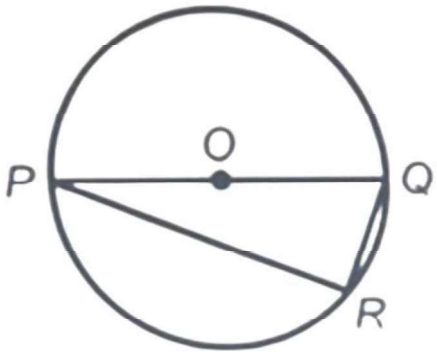
3D ACADEMY

Test #8

7) What is the square root of $25 \cdot 3^{16}$?
   A. $5 \cdot 3^4$
   B. $5 \cdot 3^8$
   C. $5 \cdot 3^{14}$
   D. $5 \cdot 3^{16}$

8) A line has a slope of -2 and passes through the point $(4, -5)$. What is the equation of this line in standard form ?
A. $2x + y = 3$
B. $2x + y = 13$
C. $2x + y = -6$
D. $2x + y = -13$

9) In the figure, $\overline{PQ}$ is a diameter of circle O, and R lies on the circle. If QR is 1 unit and the Radius of the circle is $\frac{5}{2}$ units, how many units is PR ?
   A. $\frac{\sqrt{21}}{2}$
   B. 4
   C. $\sqrt{24}$
   D. $\sqrt{26}$



3D ACADEMY

Test #8

10) A student gave this proof that $3^0 = 1$:

$3^0 = 3^{(2-2)}$	Subtraction fact
$3^{(2-2)} = \frac{3^2}{3^2}$	Quotient property of exponents
$\frac{3^2}{3^2} = 3^2 \cdot \frac{1}{3^2}$	Definition of division
$3^2 \cdot \frac{1}{3^2} = 1$	_____
$3^0 = 1$	Transitive property of equality

What is the justification for the fourth step of the proof?

A. Identity property for multiplication
B. Inverse property of multiplication
C. Identity property for addition
D. Multiplication property of equality

11) What is the standard form equation of a circle with a diameter that has endpoints of $(-1,6) and\ (5,2)$?

A. $(x-2)^2 + (y-4)^2 = 13$
B. $(x+2)^2 + (y+4)^2 = 13$
C. $(x-2)^2 + (y+4)^2 = 26$
D. $(x+2)^2 + (y-4)^2 = 26$

12) Jordan is considering purchasing two posters, Each dimension of one poster is three times the corresponding dimension of the other poster. How many times greater is the area of the larger poster than the smaller one?

A. 3
B. 6
C. 9
D. 12

3D ACADEMY

Test #8

13) If Event 1 occurs before Event 2, which of the following pairs of events are independent?

A. Event 1: Toss a coin.
Event 2: toss the same coin again.

B. Event 1: Select a card from a deck of 52 cards.
Event 2: Select another card from the same deck without replacing the first card.

C. Event 1: it begins raining outside at 1:15 P.m.
Event 2: the sidewalk outside is wet at 1:20 p.m.

D. Event 1: Select a marble from a bag that has 15 green marbles and 12 red marbles.
Event 2: shake the bag containing the remaining marbles and select a second marble.

14) Solve $2|3x-8|-3=11$

A. $x=4\ or\ x=\frac{4}{3}$

B. $x=\frac{20}{3}\ or\ x=-\frac{4}{3}$

C. $x=8\ or\ x=-\frac{8}{3}$

D. $x=5\ or\ x=\frac{1}{3}$

15) Simplify $\frac{\sqrt{8x^3}}{\sqrt{27x}}\ for\ x>0$

A. $\frac{4x}{3}$

B. $\frac{2x}{3}$

C. $\frac{2x\sqrt{6}}{9}$

D. $\frac{2x\sqrt{6}}{3}$

3D ACADEMY

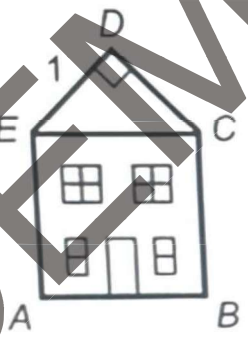
Test #8

16) Completely factor $4x^2 - 12xy + 9y^2$

A. $(2x - 3y)^2$
B. $(2x + 9y)^2$
C. $(4x - 9y)^2$
D. $4((x - 3y)^2$

17) This is a scale model of a house. Triangle EDC is as isosceles right triangle, and ABCE is a square, if $DE = 1$, what is the perimeter of $ABCDE$?

A. 5
B. 6
C. $2 + 3\sqrt{2}$
D. $4\sqrt{2}$



18) For the equation $x^2 - 2a = 0$, what value(s) of a will result in an equation with two distinct real roots?

A. $a < -2$
B. $a > 0$
C. $a = 0$
D. $a < 0$

19) The area of a triangle is found using $A = \frac{1}{2}bh$, where b is the base had h is the height. Which formula yields the height of a triangle?

A. $h = \frac{2A}{b}$
B. $h = \frac{A}{2b}$
C. $h = \frac{AB}{2}$
D. $h = \frac{b}{2A}$

3D ACADEMY

Test #8

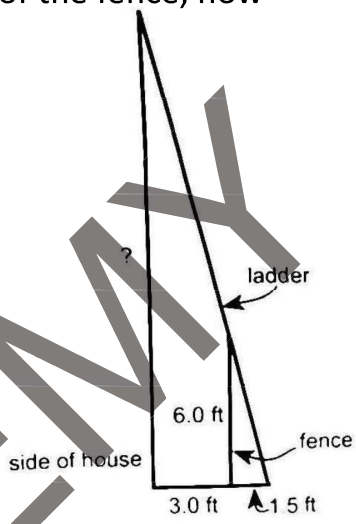
20) A ladder is leaning against the side of house. The base of the ladder is 1.50 ft from the base of a 6.00 ft tall fence. The fence is 3.00 ft from, and parallel to, the side of the house. If the ladder touches the top of the fence, how many feet does the ladder reach up the house's side?

A. 18 ft

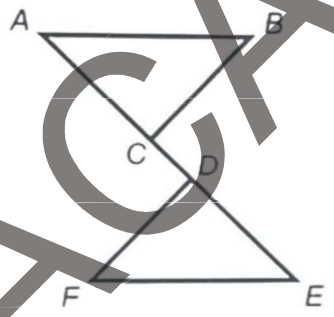
B. 19 ft

C. 20 ft

D. 21 ft



21) Given $\Delta BAC \cong \Delta FED, C \text{ and } D$ are on $\overline{AE}$, prove $\overline{AB} \parallel \overline{EF}$



Statements	Reasons
1. $\Delta BCA \cong FED$	Given
2. $\angle BAE \cong \angle FEA$	----------
3. $\overline{AB} \parallel \overline{EF}$	----------

Which is the reason for statement 3?

A. If two lines cut by a transversal form a pair of congruent alternate interior angles, then the two lines are parallel.

B. If two lines cut by a transversal form a pair of congruent alternate exterior angles, then the two lines are parallel.

C. Corresponding parts of congruent triangles are congruent.

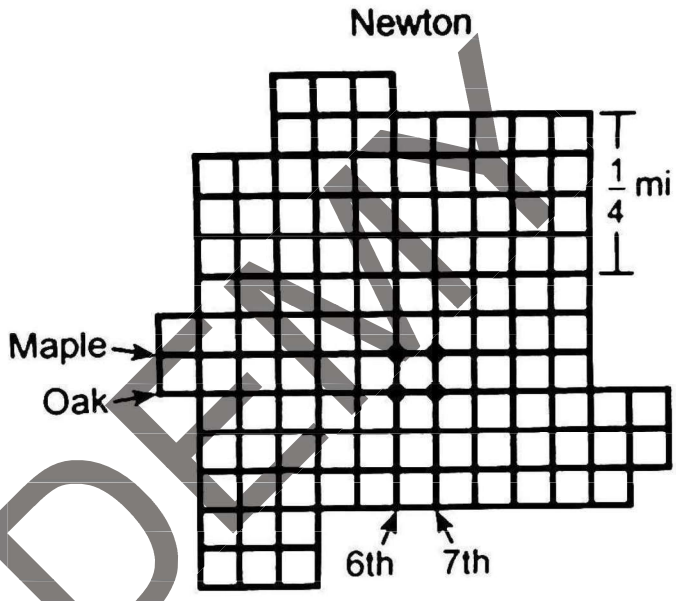
D. Congruent corresponding sides of congruent triangles are parallel.

3D ACADEMY

Test #8

22) The town of newton purchased a new emergency siren that can be heard within $\frac{1}{2}$ mile radius. Assuming the square blocks aree all the same size, at which corner should the police install the siren so that it is within $\frac{1}{2}$ mile of any street corner in newton ?

A. Maple and 6th
B. Maple and 7th
C. Oak and 6th
D. Oak and 7th



23) Jade bought 2 similar cylindrical planters. Both the diameter and height of the larger one were 3 times that of the smaller one. It took 1 bag of soil to fill the small planter. How many bags of soil does jade need to fill the larger planter?

A. 3
B. 9
C. 10
D. 27

24) Find the difference. $(x^2 + 4x + 4) - (5x^2 - x - 2)$

A. $4x^2 + 3x + 2$
B. $4x^2 + 3x + 6$
C. $-4x^2 + 5x + 6$
D. $-4x^2 + 5x + 2$

3D ACADEMY

Test #8

25) Which of the following expressions is a simplified form of this expression?
$(\sqrt{16x^6})^2(\sqrt{4x^4})^2$

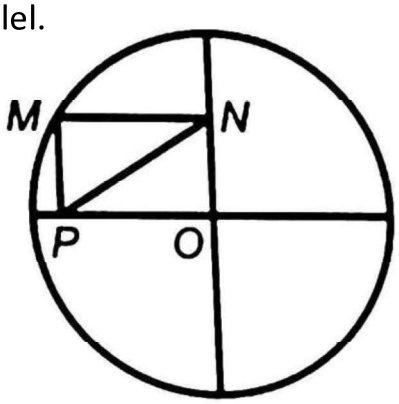
A. $8x^6$
B. $64x^{10}$
C. $64x^{24}$
D. $4{,}096x^{20}$

26) The equation $d = 16t^2$ models the distance (d), in feet, that a free-falling object travels in t seconds. According to this model, how many seconds does it take a free-falling object to travel 162 ft?

A. $\frac{9}{4}$
B. $\frac{9\sqrt{2}}{2}$
C. $\frac{9\sqrt{2}}{4}$
D. $\frac{9\sqrt{2}}{16}$

27) Circle O has a radius of 2 cm, and MNOP is a rectangle. Which reason justifies $PN = 2cm$?

A. The diagonals of a rectangle are congruent.
B. Both pairs of opposite sides of a rectangle are parallel.
C. The sum of the angles in a rectangle is $360°$.
D. Pythagorean theorem



28) If $f(x) = x^2 - 3x - 1$ find $f(-2)$.

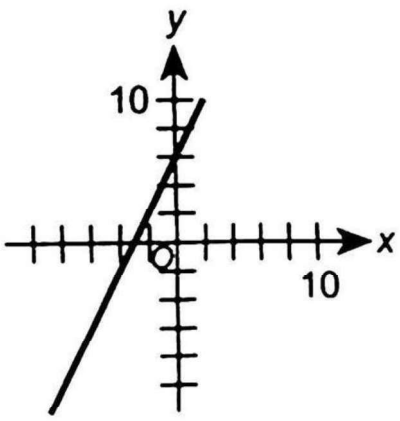
A. -11
B. -3
C. 1
D. 9

3D ACADEMY

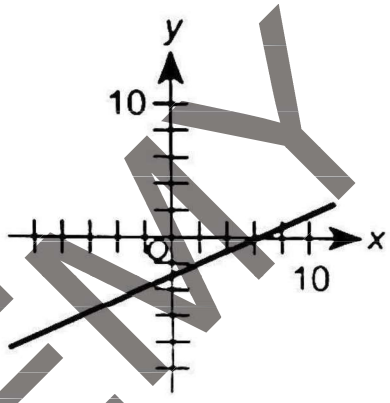
Test #8

29) A function $y = f(x)$, consists of the set of all ordered pairs such that the first member of the ordered pair is 3 less than $\frac{1}{2}$ the second member. Which of the following is the graph of this function?

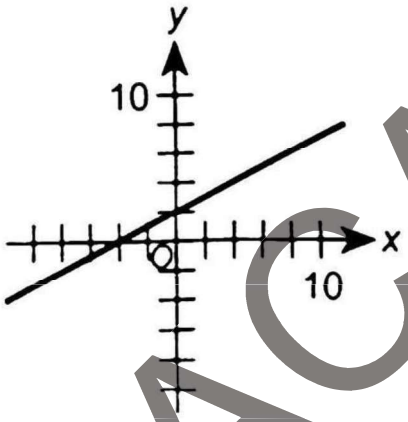
A.



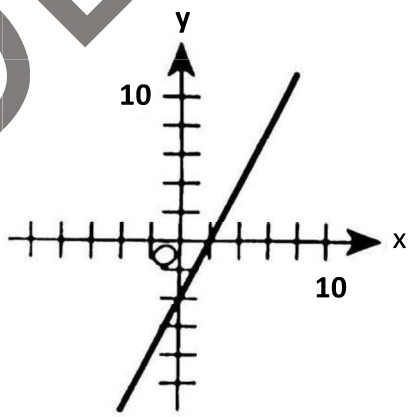
C



B



D



3D ACADEMY

Test #8

30) Which set of measurements could be the side lengths of an obtuse triangle?

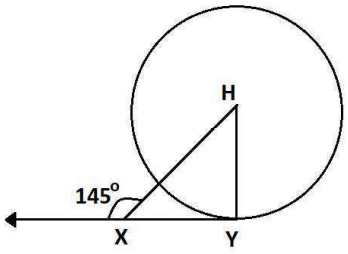
A. $3\ in, 4\ in, 4\ in.$
B. $5\ in, 12\ in, 13\ in.$
C. $6\ in\ , 7\ in, 12\ in.$
D. $6\ in, 8\ in, 9\ in.$

31) What is the equation in slope-intercept form of the line that passes through the point $(2, -2)$ and is perpendicular to the line represented by $y = \frac{2}{5}x + 2$?

A. $y = \frac{5}{2}x - 7$
B. $y = \frac{5}{2}x + 7$
C. $y = -\frac{5}{2}x + 3$
D. $y = -\frac{5}{2}x - 3$

32) in the circle, H is the center, and $\overline{XY}$ is tangent to the circle, what is the measure of $\angle XHY$?

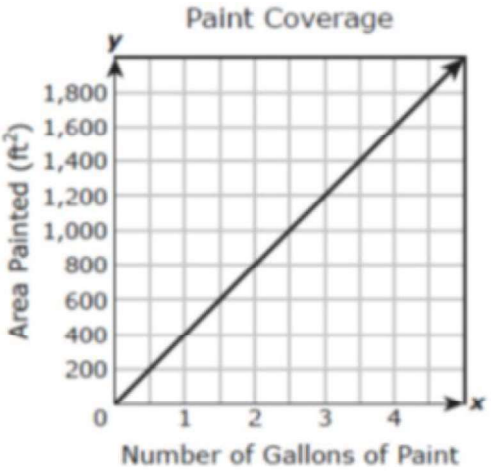
A. $25°$
B. $55°$
C. $90°$
D. $125°$



33) The graph shows the linear relationship between the maximum area in square feet that can be painted and the number of gallons of paint used.

Which of these represents the rate of change of the maximum area painted with respect to the number of gallons of paint used?

A. $200\ ft^2/gal$
B. $\frac{1}{200}\ ft^2/gal$
C. $400 ft^2/gal$
D. $\frac{1}{400}\ ft^2/gal$



3D ACADEMY

Test #8

34) Two customers went to a post office to buy postcards and large Envelops. Each postcard costs the same amount, and a large envelope costs the same amount.

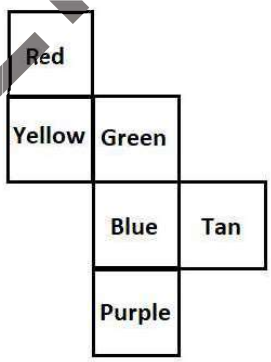
- The first customer paid $12 for 14 postcards and 5 large envelops.
- The second customer paid $24.80 for 10 postcards and 15 large envelops.

What was the cost in dollars of each large envelops?

A. $1.42
B. $0.35
C. $1.15
D. $0.63

35) Imagine that this net will be folded into a cube. Which color will not be adjacent to the red face?

A. Purple
B. Green
C. Blue
D. Tan



36) Which function is equivalent to $f(x) = -4(x+7)^2 - 6$?

A. $f(x) = -4x^2 - 56x - 202$
B. $f(x) = -4x^2 + 14x + 43$
C. $f(x) = -4x^2 - 56x - 172$
D. $f(x) = -4x^2 + 190$

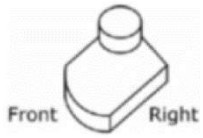
37) Which expression is a factor of $21x^2 + 13x - 20$?

A. $3x - 4$
B. $7x - 5$
C. $7x + 4$
D. $3x + 5$

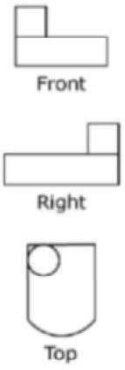
3D ACADEMY

Test #8

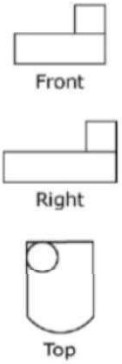
38) Which set shows the front, right, and top views of this three-dimensional figure?



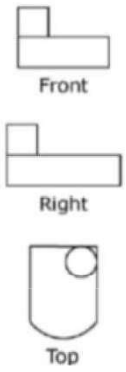
A



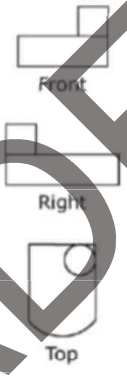
C



B



D

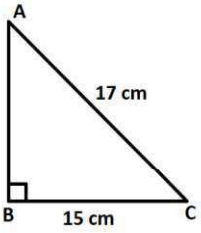


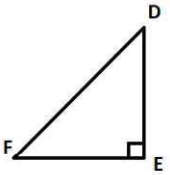
39) In a sequence of numbers $a_3 = 0$, $a_4 = 6$, $a_5 = 12$, $a_6 = 18$, $and\ a_7 = 24$. Based on this information, which equation can be used to find n^{th} term in the sequence, a_n?

A. $a_n = -6n + 18$
B. $a_n = -18n + 6$
C. $a_n = 6n - 18$
D. $a_n = 18n - 6$

40) Kayle was cutting right triangles from wood to one of an art project. Two of the right triangles she cut are shown below. If $\Delta ABC \sim \Delta DEF$ with right angle B and E, BC=15 cm and AC= 17 cm. what is the measure of $\angle F$ to the nearest degree?

A. 28°
B. 41°
C. 62 °
D. 88°





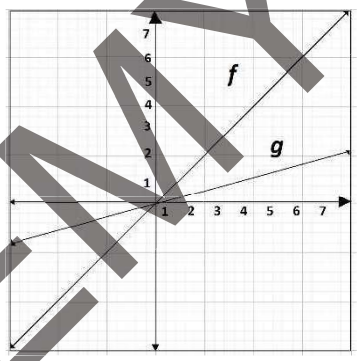
3D ACADEMY

Test #8

41) Which equation represents the line passing through the points $(3, 2)$ and $(-9, 6)$?

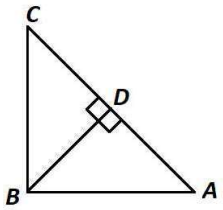
A. $x - 3y = 9$
B. $x + 3y = 9$
C. $3x - y = -9$
D. $3x + y = 9$

42) The graph of linear function f *and* g are shown on the grid
Which function is best represented by the graph of g?



A. $g(x) = f(x) - 4$
B. $g(x) = \frac{1}{3}f(x)$
C. $g(x) = f(x) - 2$
D. $g(x) = 3f(x)$

43) In the diagram below of ΔABC, $\angle ABC$ is a right angle $AC = 12, AD = 8$ and altitude $\overline{BD}$ is drown, what is the length of $\overline{BC}$?



A. $4\sqrt{2}$
B. $4\sqrt{3}$
C. $4\sqrt{5}$
D. $4\sqrt{6}$

44) A circle whose center is the origin passes through the point $(-5,12)$ which point also lies on the circle?

A. $(10, 3)$
B. $(-12, 13)$
C. $(11, 2\sqrt{12})$
D. $(-8, 5\sqrt{21})$

3D ACADEMY

Test #8

45) What is the contrapositive of the statement below?
"If today is Friday, then tomorrow is Saturday"

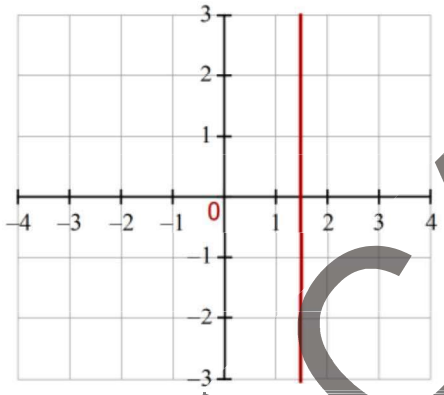
A. If tomorrow is not Saturday, them today is not Friday.
B. If today is Saturday, then tomorrow is not Friday.
C. If tomorrow is Saturday, then today is Friday.
D. If today is not Friday, then tomorrow is not Saturday.

46) Which statement about the graph of $y = 8(0.25)^x$ *is true*?

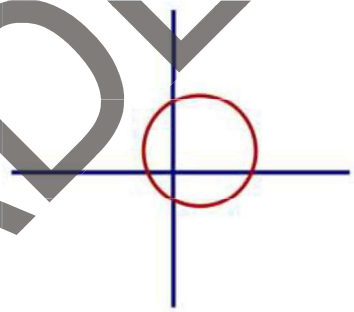
A. The coordinates of the x-intercept are $(0.25, 0)$
B. The coordinates of the y-intercept are $(0\ , 8)$
C. The equation of the asymptote is $x = 0$
D. The graph includes the point $(2, 1)$

47) Which graph represents a function?

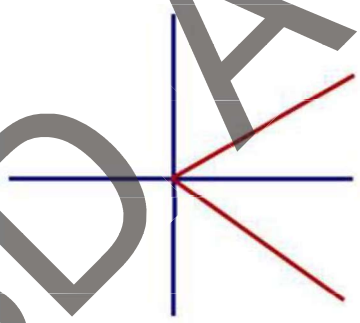
A)



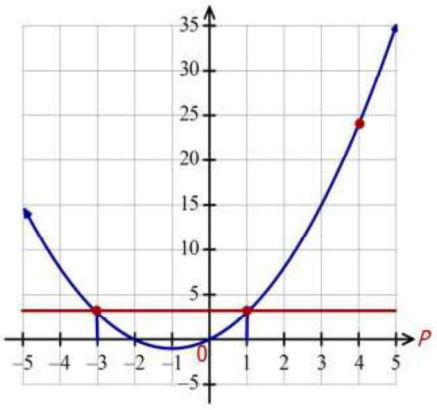
B)



C)



D)



3D ACADEMY

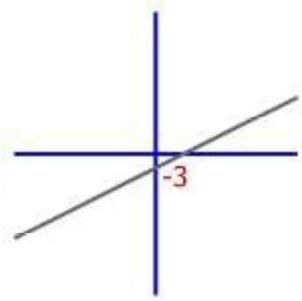
Test #8

48) Two semicircles are congruent and are collinear. How many points and lines of symmetry do they have, if any ?

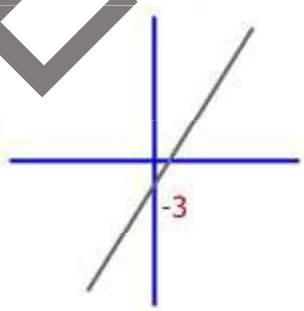
A) 1 Point, 2 line
B) 2 point, 1 line
C) 1 point, 1 line
D) 0 point, 0 line

49) In $f(x) = y$, a point with coordinates x, y lies. If the first member of the coordinates lies 3 less than the second member, which graph represents $f(x)$?

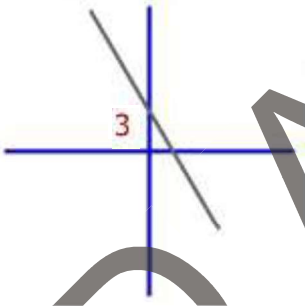
A)



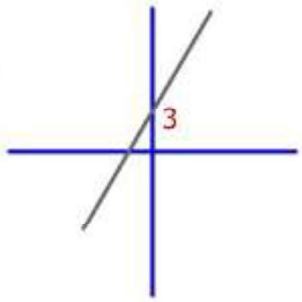
B)



C)



D)



50) Triangle JGR is similar to triangle MST . which statement is not always true?

A) $\angle J \cong \angle M$
B) $\angle G \cong \angle T$
C) $\angle R \cong \angle T$
D) $\angle G \cong \angle S$

3D ACADEMY

Test #8

Answer Keys:

Question	Answer	Question	Answer
1	D	26	C
2	B	27	A
3	C	28	D
4	C	29	A
5	B	30	C
6	C	31	C
7	B	32	B
8	A	33	C
9	C	34	A
10	B	35	C
11	A	36	A
12	C	37	B
13	A	38	D
14	D	39	C
15	C	40	A
16	A	41	B
17	C	42	B
18	B	43	B
19	A	44	C
20	A	45	A
21	A	46	B
22	A	47	D
23	D	48	A
24	C	49	D
25	B	50	B

3D ACADEMY